

BEYOND NOMINAL MARKERS: NUISANCE-AWARE PANEL SELECTION FOR CELL-FRACTION ESTIMATION

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ABSTRACT

Estimating cell-type fractions from bulk expression can help characterize heterogeneous biological samples, but compact marker panels may confound cell abundance with technical variation and background-composition changes. We develop a nuisance-aware qPCR-style panel-selection framework that ranks candidates using discovery cells only, selects normalization references separately, and chooses panels of at most five genes by optimizing cross-validated error across nominal and simulated regimes with scaling, assay-efficiency variation, dropout, ambient RNA, background-composition changes, and their combination. Using independently generated holdout pseudobulks from held-out cells, we compare this robust procedure with strict nominal-only selection. On nominal holdout data, the robust panel achieved macro ROC AUC 0.9860 and MAE 0.0606, compared with ROC AUC 0.9865 and MAE 0.0613 for nominal-only selection. Across the seven tested regimes, robust selection reduced macro worst-case MAE from 0.1196 to 0.1103; in the combined regime, it reduced MAE from 0.1196 to 0.1103 and increased ROC AUC from 0.9044 to 0.9321. These results show lower quantitative error for nuisance-aware selection under the specified simulated perturbations, but do not establish statistical significance, universal robustness, or transfer to experimentally measured qPCR data.

1 INTRODUCTION

Biological samples commonly contain mixtures of cell populations whose relative abundances influence the interpretation of tissue state, immune infiltration, and treatment response. Estimating these abundances from bulk expression is therefore an important problem in computational biology. Reference-based methods estimate cellular composition from expression signatures (Racle et al., 2017; Aran et al., 2017), while marker-based methods and benchmarking studies emphasize the roles of reference quality, normalization, marker specificity, and mixture composition (Étienne Becht et al., 2016; Cobos et al., 2020). Targeted quantitative polymerase chain reaction (qPCR) panels offer a complementary option when a smaller, more operationally practical assay is preferred. The goal in this setting is to select a compact marker set that estimates cell fractions from normalized expression under changing measurement conditions and background composition.

Compact-panel design is difficult because marker signals can change for reasons unrelated to the target fraction. Global scaling, gene-wise assay-efficiency variation, dropout, ambient RNA, and changes in non-target composition can all alter measured expression, allowing a panel optimized on nominal mixtures to rely on relationships that fail under perturbation. Additional markers may provide complementary information but increase assay burden and can introduce redundancy, while reference genes are needed for normalization but should remain distinct from target markers.

We develop a nuisance-aware framework for selecting qPCR-style panels with at most five markers per target cell type. Candidates are ranked using discovery cells only, reference genes are selected separately by expression stability, and a fixed cell-level discovery-holdout split prevents held-out cells from influencing selection. Forward selection uses discovery pseudobulks from nominal and specified perturbed regimes and optimizes a cross-validated objective combining mean error, worst-regime error, and panel size. The perturbations include global scaling, gene-wise assay-efficiency variation, dropout, ambient RNA, alternative background composition, and their combination.

Our contributions are:

- We formulate a compact-panel objective that combines mean error across specified nuisance regimes, worst-regime error, and panel size.
- We separate discovery-only candidate ranking, normalization-reference selection, model fitting, and holdout evaluation to limit information leakage.
- We define a pseudobulk stress test with independently generated discovery and holdout mixtures covering technical perturbations, background-composition changes, and their combination.
- We evaluate both detection and quantitative estimation using discrimination, operating-point, error, correlation, calibration, and abundance-stratified metrics.
- We compare nuisance-aware selection with a strict nominal-only procedure and a best single-marker comparator on the same held-out regimes.

We test whether nuisance-aware selection lowers quantitative error across the specified holdout regimes without sacrificing nominal performance. Discovery cells are used for candidate ranking, reference selection, panel selection, and fitting; independently generated mixtures from held-out cells are reserved for evaluation. The final comparison evaluates nuisance-aware and strict nominal-only selection on the same nominal and perturbed holdout regimes, alongside a best single-marker comparator.

This study is a controlled, simulation-based evaluation of panel-selection objectives, not a claim of universal robustness to laboratory variation. The nuisance distributions are explicit stress-test conditions, and the qPCR-style scores are proxies computed from expression values rather than measured cycle-threshold values. Transfer to laboratory assays, independent biological samples, technical replicates, and empirically calibrated nuisance distributions remains to be tested.

2 RELATED WORK

Reference-based deconvolution methods estimate cell-population abundance from mixed transcriptomic measurements using multi-gene signatures. EPIC accounts for cell-type-specific mRNA content when estimating cancer and immune fractions (Racle et al., 2017); xCell applies reference signatures and spillover compensation to related immune and stromal populations (Aran et al., 2017); MCP-counter quantifies immune and stromal populations from marker signatures (Étienne Becht et al., 2016); and ImmuCellAI estimates immune populations, including T-cell subsets, from gene sets (Miao et al., 2020). These methods address the same broad estimation problem, but they assume access to transcriptome-scale measurements and score or deconvolve many genes rather than selecting a small assay. Their accuracy also depends on reference quality and coverage: biological and technical biases in reference datasets or basis matrices can affect mixture estimates, and systematic evaluations report method-dependent performance across cell types and settings (Vallania et al., 2018; Cobos et al., 2020; Sturm et al., 2019). Because the present experiment restricts observations to at most five target markers plus separate reference genes, these methods are related alternatives but are not directly comparable baselines here.

Single-cell-informed methods address reference mismatch through different modeling assumptions: Bisque learns transformations between single-cell references and bulk measurements (Jew et al., 2020), MuSiC uses multi-subject single-cell references (Wang et al., 2019), and SCDC combines multiple single-cell references (Dong et al., 2019). FARDEEP instead uses outlier-resistant fitting for tumor-expression profiles (Hao et al., 2019). These methods modify transcriptome-wide deconvolution to accommodate reference, batch, or outlier effects; our method instead selects a compact panel using discovery cells and prespecified perturbation regimes, while holding out cells for final evaluation. Thus, the comparison is methodological rather than a benchmark comparison: the present experiments test panel-selection objectives and do not evaluate transcriptome-wide deconvolution methods under the same feature constraint.

3 BACKGROUND

Quantitative polymerase chain reaction (qPCR) measures a selected set of transcripts and is therefore compatible with compact assays. Relative quantification requires normalization against reference

genes, and standard-curve- or efficiency-based procedures make different assumptions about transforming and comparing expression measurements (Larionov et al., 2005). The MIQE guidelines emphasize assay quality, amplification efficiency, normalization, and technical variability (Bustin et al., 2009); geometric averaging of multiple internal controls is a standard normalization strategy (Vandesompele et al., 2002). These considerations motivate selecting normalization references separately from target markers and treating assay-related variation as nuisance variation.

Cell-fraction estimation from mixed expression measurements is related to reference-based deconvolution, which infers population abundances from cell-type-specific expression signatures. Existing methods use transcriptome-scale measurements and differ in their treatment of references, normalization, and mixture composition (Racle et al., 2017; Aran et al., 2017; Étienne Becht et al., 2016; Cobos et al., 2020). Our setting instead assumes a compact targeted assay: a small marker panel and separate normalization references are selected from discovery cells, then evaluated under controlled observation and background-composition perturbations.

3.1 PROBLEM SETTING

Let $\mathcal{G}$ be the gene set and $\mathcal{C}$ the set of population labels. We observe a nonnegative single-cell expression matrix $X \in \mathbb{R}_{\geq 0}^{N \times |\mathcal{G}|}$ and labels $c_i \in \mathcal{C}$ for $i = 1, \dots, N$. For target population $t \in \mathcal{C}$, the goal is to estimate its fraction $y_t \in [0, 1]$ in a mixed sample. We use a one-versus-rest formulation: cells labeled t are the target, and cells labeled $\mathcal{C} \setminus \{t\}$ form a heterogeneous background. This formulation is applied independently to each target.

For target t , let $M_t \subset \mathcal{G}$ be a marker panel with $1 \leq |M_t| \leq 5$, and let $R \subset \mathcal{G}$ be a disjoint normalization-reference set. For mixed expression vector $z \in \mathbb{R}_{\geq 0}^{|\mathcal{G}|}$, define

$$s_{M_t, R}(z) = \frac{1}{|M_t|} \sum_{g \in M_t} z_g - \frac{1}{|R|} \sum_{g \in R} z_g.$$

Because z contains expression values rather than physical cycle-threshold measurements, $s_{M_t, R}$ is a delta-expression proxy, not a direct C_t or ΔC_t measurement. A univariate linear model maps this score to the target fraction, with predictions clipped to $[0, 1]$.

A simulated pseudobulk contains K sampled cells. We draw an integer target count n_t with $1 \leq n_t \leq K - 1$, allocate the remaining cells across background populations with a Dirichlet-multinomial draw, and average the sampled expression vectors. Its known target fraction is

$$y_t = \frac{n_t}{K}.$$

This construction treats observed single-cell profiles as empirical representatives of their labeled populations and approximates a mixed sample by averaging sampled profiles, analogous to pseudobulk strategies that aggregate cells to reduce cell-level pseudoreplication and support sample-level inference across biological conditions or replicates (Thurman et al., 2021; Squair et al., 2021; Crowell et al., 2020). It assumes that these profiles and the specified mixture model adequately proxy the mixtures used for evaluation.

Let η index a nuisance regime and T_η denote its mixture-generation or observation operator. The regimes cover nominal measurements, global scaling, gene-wise assay efficiency, dropout, ambient RNA, alternative background composition, and their combination. Observation perturbations modify measured expression without changing the known target fraction; background-composition variation changes the non-target cells contributing to the mixture while preserving the target count. Discovery and holdout nuisance parameters are sampled independently from fixed ranges using separate seeds.

Candidates and references are determined from discovery cells only. For candidate panel M , robust selection minimizes

$$J(M) = \frac{1}{|\mathcal{E}|} \sum_{\eta \in \mathcal{E}} \text{MAE}_\eta(M) + \lambda \max_{\eta \in \mathcal{E}} \text{MAE}_\eta(M) + \gamma |M|,$$

where $\mathcal{E}$ is the set of discovery regimes, $\lambda = 0.50$, and $\gamma = 0.001$ in the final configuration. Each regime-specific error uses five-fold cross-validated linear regression. Forward selection stops at five

markers or when the objective decrease is below 0.002. The nominal-only comparator instead selects and fits using nominal discovery pseudobulks alone; both procedures are evaluated on the same independently generated holdout regimes.

4 METHOD

4.1 DISCOVERY AND HOLDOUT PARTITION

We partition each Leiden cluster into discovery and holdout subsets using random seed 42. The discovery subset contains the rounded 70% fraction, subject to retaining at least two cells in each subset. The same partition is used for every target. Marker ranking, reference-gene selection, pseudobulk generation, and model fitting use discovery cells; holdout cells are used only for final evaluation.

4.2 DISCOVERY-ONLY CANDIDATE AND REFERENCE SELECTION

For each target cluster, we use a Wilcoxon rank-sum test to compare that cluster with all other discovery cells, following standard single-cell differential-expression workflows (Luecken & Theis, 2019). The first 30 returned genes form the target-specific candidate list. Candidate ranking is performed independently for each target and excludes holdout cells.

For each gene g , we compute the discovery-cell stability score

$$\frac{\text{sd}_c(\bar{x}_{c,g})}{|\bar{x}_{\text{all},g}| + 10^{-6}},$$

where $\bar{x}_{c,g}$ is the mean expression in cluster c and $\bar{x}_{\text{all},g}$ is the overall discovery-cell mean. We exclude genes expressed in fewer than 20% of discovery cells and genes in any target-specific candidate list. The 200 lowest-scoring genes form the reference pool; its first three genes define R , which is fixed before marker selection.

4.3 PSEUDOBULK GENERATION

For each target, each pseudobulk contains $K = 100$ cells. A target fraction is sampled from $[0.01, 0.10]$, $[0.10, 0.50]$, or $[0.50, 0.90]$ with probabilities 0.30, 0.45, and 0.25, respectively; the rounded target-cell count is clipped to $[1, K - 1]$. The remaining cells are assigned to non-target clusters by a Dirichlet draw followed by a multinomial draw. Cells are sampled with replacement from the appropriate discovery or holdout subset, their expression vectors are averaged, and the known fraction is $y_t = n_t/K$.

For each target, the nominal-only procedure uses 200 nominal discovery pseudobulks. Robust selection uses 200 discovery pseudobulks for each of the seven regimes: nominal, global scaling, assay efficiency, dropout, ambient RNA, background composition, and combined perturbations. Holdout evaluation uses 300 pseudobulks per regime. Discovery and holdout mixtures are generated from their respective cell subsets using separate seed offsets; the holdout mixtures are not used during selection or fitting.

4.4 NUISANCE REGIMES

Let η index a regime and T_η denote its sampling and transformation procedure. The regimes are nominal, global scaling, gene-wise assay efficiency, dropout, ambient RNA, alternative background composition, and a combined regime. Scaling and assay efficiency multiply expression values; dropout sets selected values to zero; ambient RNA mixes the pseudobulk with a randomly selected background-cell profile; and alternative background composition changes the Dirichlet concentration used to allocate non-target cells. The parameter ranges are $[0.70, 1.30]$ for scaling and assay efficiency, $[0.05, 0.25]$ for dropout, $[0.05, 0.20]$ for ambient contamination, and $[0.20, 3.00]$ for the background concentration. The combined regime applies all five perturbations, with independently seeded discovery and holdout generation.

4.5 NORMALIZED PANEL SCORES

For each pseudobulk vector z , we compute

$$s_{M_t, R}(z) = \frac{1}{|M_t|} \sum_{g \in M_t} z_g - \frac{1}{|R|} \sum_{g \in R} z_g.$$

This expression-based delta score is a proxy rather than a physical cycle-threshold or ΔC_t measurement. A univariate ordinary least-squares regression maps the score to y_t , and predictions are clipped to $[0, 1]$.

4.6 PANEL SELECTION

Using nominal discovery pseudobulks only, we select a target-specific panel by forward search over the ranked candidates. We initialize the panel with the candidate having the lowest five-fold cross-validated MAE, then greedily append the candidate giving the lowest nominal MAE. An addition is accepted only when it improves MAE by at least 0.002, and selection stops at five markers or when no candidate meets that criterion. The initial candidate is retained as the single-marker comparator.

For robust selection, each candidate panel is evaluated in every discovery regime with shuffled five-fold cross-validation using random state 42. Let $\text{MAE}_\eta(M)$ be the cross-validated MAE in regime η , and let $\mathcal{E}$ contain the seven regimes. We minimize

$$J(M) = \frac{1}{|\mathcal{E}|} \sum_{\eta \in \mathcal{E}} \text{MAE}_\eta(M) + \lambda \max_{\eta \in \mathcal{E}} \text{MAE}_\eta(M) + \gamma |M|,$$

with $\lambda = 0.50$ and $\gamma = 0.001$. Forward selection starts with the candidate minimizing $J(M)$ and adds the candidate producing the smallest objective, stopping at five markers or when the decrease is below 0.002. The robust panel therefore uses all seven discovery regimes, while the nominal comparator uses only nominal discovery pseudobulks.

4.7 FINAL FITTING AND EVALUATION

After selection, the nominal panel and single-marker model are fitted using nominal discovery pseudobulks only. The robust model is fitted to the concatenation of the discovery pseudobulks from all seven regimes. Each fitted model is applied without refitting to the target-specific holdout regimes generated from held-out cells, allowing the two panel-selection strategies to be compared under identical holdout conditions.

For detection, a pseudobulk is positive when $y_t \geq 0.35$. ROC AUC uses the continuous predictions clipped to $[0, 1]$. We also report the maximum sensitivity over prediction thresholds with specificity at least 0.90, and specificity after thresholding predictions at 0.35. Quantitative metrics are MAE, RMSE, R^2 , Spearman and Pearson correlation, calibration slope and intercept, and MAE for $y_t \leq 0.10$ or $y_t \geq 0.50$. Metrics are computed per target cluster and summarized by an unweighted macro mean.

5 EXPERIMENTAL SETUP

5.1 DATASET AND CELL COMPOSITION

We use the supplied single-cell expression dataset and its Leiden cluster labels. For each target cluster, cells assigned to that cluster define the target population, while all other labeled cells define the heterogeneous background. The same labels are used for the discovery-holdout split, marker ranking, pseudobulk generation, and target-wise evaluation.

Figure 1 shows a UMAP embedding of the labeled cells (McInnes et al., 2018). The embedding is descriptive only: it is not used for marker selection, pseudobulk generation, model fitting, or metric calculation.

5.2 EVALUATION PROTOCOL

Within each Leiden cluster, approximately 70% of cells are assigned to discovery and the remainder to holdout, with at least two cells retained in each subset. Marker ranking and reference-gene selection

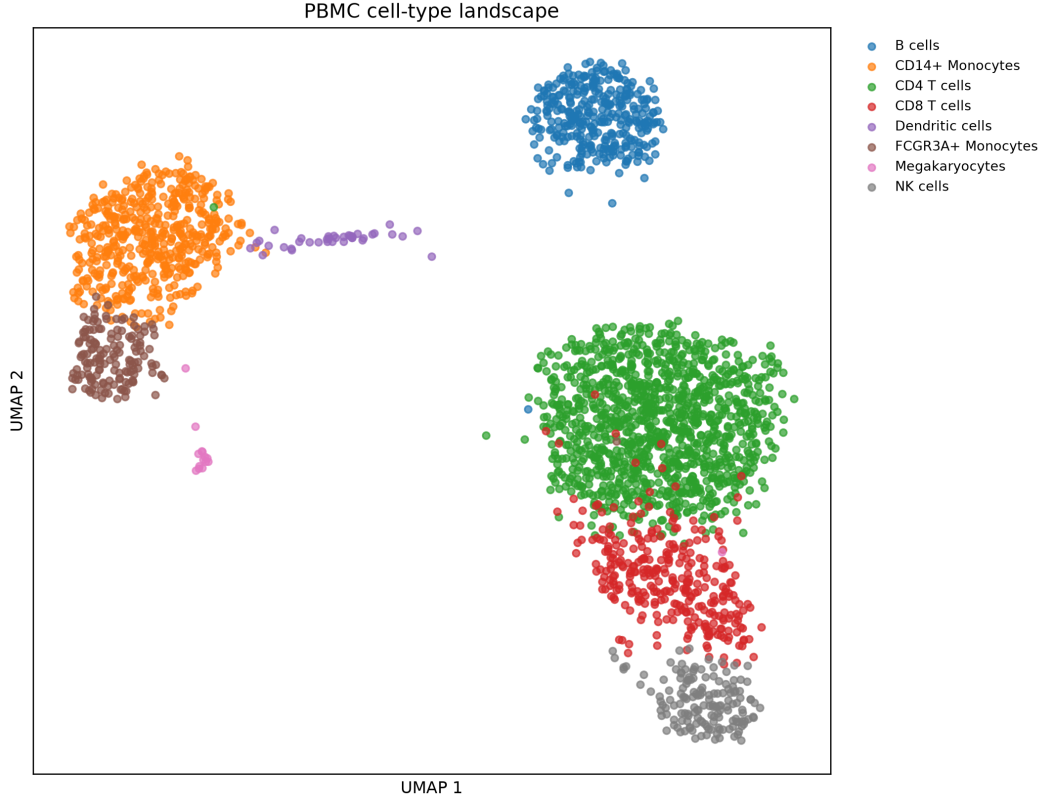


Figure 1: UMAP embedding of the supplied single-cell dataset, colored by Leiden cluster. The labeled clusters define the target and background populations used for marker discovery and pseudobulk simulation.

use discovery cells only. For each target and regime, we generate 200 discovery pseudobulks and 300 holdout pseudobulks. Nominal-only selection and fitting use only nominal discovery pseudobulks; robust selection and fitting use discovery pseudobulks from all seven regimes. Holdout pseudobulks are generated from held-out cells and are not used during selection or fitting.

The experiment uses 42 as the base random seed, with separate seed offsets for target clusters and regimes. Each pseudobulk contains 100 cells sampled with replacement. A target fraction is sampled from $[0.01, 0.10]$, $[0.10, 0.50]$, or $[0.50, 0.90]$ with probabilities 0.30, 0.45, and 0.25, respectively; after rounding, the target-cell count is clipped to $[1, 99]$. The remaining cells are allocated among background clusters by a Dirichlet draw followed by a multinomial draw. The selected expression vectors are averaged, and the known target fraction is the target-cell count divided by 100.

The seven regimes are nominal, global scaling, assay efficiency, dropout, ambient RNA, background composition, and combined. Global scaling and gene-wise assay-efficiency factors are sampled from $[0.70, 1.30]$; dropout rates from $[0.05, 0.25]$; ambient-RNA fractions from $[0.05, 0.20]$; and background-composition Dirichlet concentrations from $[0.20, 3.00]$. Scaling and assay efficiency multiply expression values, dropout sets selected values to zero, ambient RNA mixes each pseudobulk with a randomly selected background-cell profile, and background-composition variation changes the Dirichlet concentration used to allocate non-target cells. The combined regime applies all five perturbations. Discovery and holdout nuisance parameters are sampled independently using separate seed offsets.

5.3 MODELS AND HYPERPARAMETERS

For each target cluster, the first 30 genes returned by the discovery-only Wilcoxon ranking form the marker-candidate list. Reference genes are ranked separately by stability of cluster-level mean

expression, after excluding genes that occur in any target-specific candidate list and genes expressed in fewer than 20% of discovery cells. The first three genes from the resulting stability-ranked pool are used as normalization references. For a marker panel M and reference set R , the model feature is the mean expression of M minus the mean expression of R . A univariate ordinary least-squares linear regression maps this feature to the known target fraction, and predictions are clipped to $[0, 1]$.

The nominal-only panel is selected by greedy forward search using five-fold cross-validated MAE on nominal discovery pseudobulks. The robust panel is selected by minimizing mean regime-specific cross-validated MAE plus a worst-regime penalty with $\lambda = 0.50$ and a panel-size penalty with $\gamma = 0.001$. Both searches consider at most five markers and add a marker only when the relevant objective decreases by at least 0.002. The first marker selected by nominal-only search is evaluated as the single-marker comparator. Cross-validation uses five shuffled folds with random state 42; holdout metrics are not used for selection or hyperparameter choice.

5.4 EVALUATION METRICS

A pseudobulk is labeled positive for detection when its known target fraction is at least 0.35. ROC AUC is computed from the continuous clipped fraction predictions. We also report the maximum sensitivity among thresholds attaining specificity of at least 0.90, and specificity after thresholding predictions at 0.35. Each quantity is calculated separately for every target cluster and summarized using an unweighted macro mean.

For quantitative estimation, we report MAE, RMSE, R^2 , Spearman correlation, Pearson correlation, calibration slope, and calibration intercept, together with MAE for true fractions at most 0.10 and at least 0.50. For each target, worst-case MAE is the maximum across the seven holdout regimes; macro worst-case MAE is the unweighted mean of these target-wise maxima. Regime-specific metrics are averaged across target clusters without weighting by the number of cells or pseudobulks. The experiments report no uncertainty estimates or statistical tests.

6 RESULTS

On nominal holdout pseudobulks, the robust panel achieved macro ROC AUC 0.9860, MAE 0.0606, RMSE 0.0792, Spearman correlation 0.9491, Pearson correlation 0.9642, and R^2 0.8886. Its low- and high-fraction MAEs were 0.0484 and 0.0867, respectively. The strict nominal-only panel had ROC AUC 0.9865 and MAE 0.0613, while the best single-marker comparator had ROC AUC 0.9850 and MAE 0.0667. Thus, the robust panel had lower nominal MAE than either comparator and lower nominal RMSE than the single-marker comparator, while the nominal-only panel had the highest ROC AUC among the two multi-gene panels.

The selected-marker expression profile and cluster-wise ROC curves provide complementary views of marker specificity and nominal detection performance (Figures 3 and 4). These diagnostics describe the final panel on the simulated nominal holdout data and do not establish performance for experimentally measured qPCR assays.

The nominal predicted-versus-true fractions show the relationship between estimated and known target fractions across abundance bins and target clusters (Figure 5). The corresponding residuals show how prediction error varies with abundance and target cluster (Figure 6). These plots characterize the simulated nominal setting; they do not establish calibration for experimentally measured qPCR cycle-threshold values.

The strict ablation compares a nominal-only panel, selected and fitted using nominal discovery pseudobulks, with a robust panel selected and fitted using nuisance-augmented discovery pseudobulks. Robust selection reduced macro worst-case MAE from 0.1196 to 0.1103, an absolute difference of 0.0093. On nominal holdout pseudobulks, the robust and nominal-only panels had MAEs of 0.0606 and 0.0613 and ROC AUC values of 0.9860 and 0.9865, respectively. The robust panel therefore had lower worst-case and nominal MAE in this comparison, but not the highest nominal ROC AUC.

In the final protocol, robust selection had lower macro MAE than nominal-only selection in all seven holdout regimes. The robust-minus-nominal MAE differences were -0.0007 for nominal data, -0.0029 under global scaling, -0.0056 under assay-efficiency variation, -0.0061 under dropout, -0.0074 under ambient RNA, -0.0016 under background-composition variation, and -0.0093

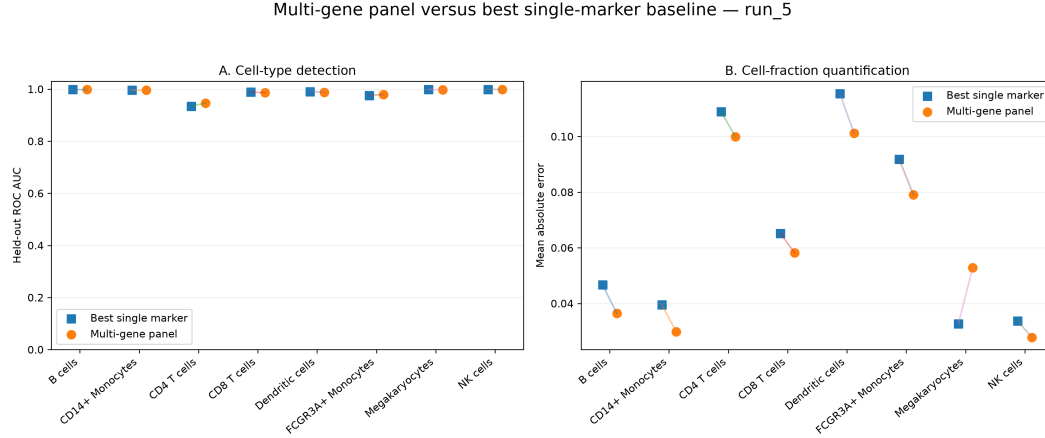


Figure 2: Per-cluster nominal holdout comparison between the selected multi-gene panel and the best single-marker comparator in the final protocol. Connected points show paired ROC AUC and MAE values for each target cluster.

under the combined regime. The combined regime was the most difficult for both methods: robust and nominal-only MAE were 0.1103 and 0.1196, respectively, while ROC AUC was 0.9321 and 0.9044. Sensitivity at fixed specificity was 0.8146 for robust selection and 0.7872 for nominal-only selection; specificity at the 0.35 threshold was similar (0.9575 and 0.9571). Detection results were mixed: nominal-only ROC AUC was slightly higher under nominal data, global scaling, and ambient RNA, whereas robust selection had higher ROC AUC under assay-efficiency variation, dropout, background-composition variation, and the combined regime.

The regime-specific ROC curves and predicted-versus-true plots summarize detection and quantitative estimation across the tested conditions (Figures 8 and 9). For robust selection, macro MAE was highest under the combined regime (0.1103), followed by dropout (0.0857) and ambient RNA (0.0821), and lowest under background-composition variation (0.0599). Robust-panel ROC AUC ranged from 0.9321 in the combined regime to 0.9860 in the nominal regime. These results characterize performance under the specified perturbations and do not establish robustness to untested nuisance distributions or laboratory assay conditions.

Within the completed sensitivity analysis, reducing the worst-regime penalty from $\lambda = 0.50$ to $\lambda = 0.25$ reduced the mean selected panel size from 2.75 to 2.375 markers, but increased macro MAE from 0.0606 to 0.0624, RMSE from 0.0792 to 0.0806, low-fraction MAE from 0.0484 to 0.0513, high-fraction MAE from 0.0867 to 0.0903, and macro worst-case MAE from 0.1103 to 0.1112. Macro ROC AUC changed from 0.9860 to 0.9862. Thus, within these two tested configurations, the smaller panels did not offset the increase in quantitative error; the comparison is a hyperparameter sensitivity analysis rather than an estimate of variation across independent repetitions.

Taken together, the final ablation shows lower macro MAE for nuisance-aware selection than for strict nominal-only selection in each of the seven tested regimes, including the combined condition. Nominal quantitative performance was similar, while nominal ROC AUC was slightly lower for the robust panel. The experiments provide no uncertainty estimates or statistical tests, use pseudobulks from a single supplied dataset rather than independent biological samples, and do not establish performance beyond the specified perturbations. The normalized expression score should therefore be interpreted as a qPCR-style proxy rather than a direct cycle-threshold measurement.

7 CONCLUSIONS AND FUTURE WORK

We developed a nuisance-aware procedure for selecting qPCR-style marker panels of at most five genes for cell-fraction estimation from single-cell-derived pseudobulk expression. The procedure combines discovery-only marker ranking, separate normalization-reference selection, a fixed discovery-holdout split, and a multi-regime objective that penalizes mean error, worst-regime error, and panel

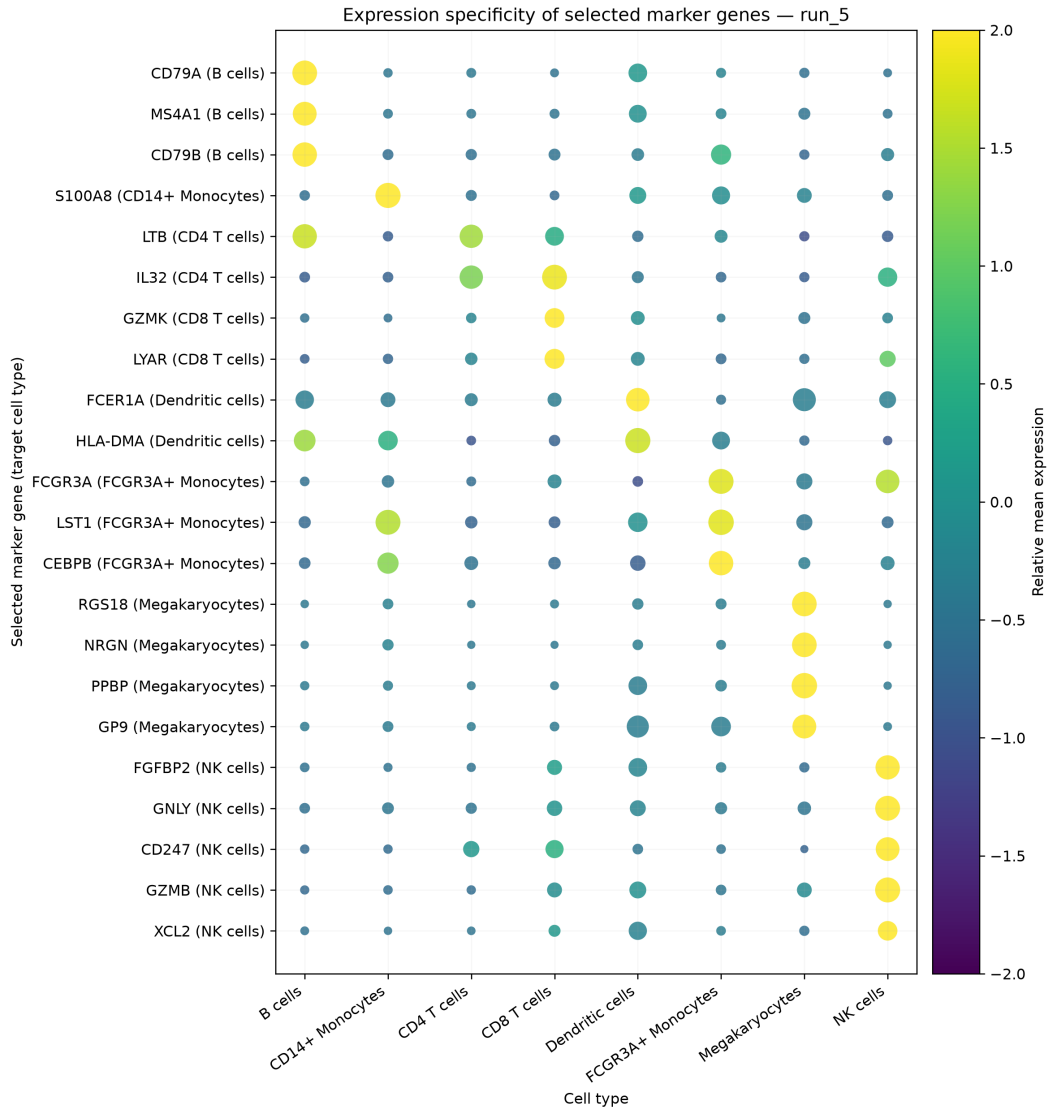


Figure 3: Expression prevalence and relative mean expression of markers selected for the final robust panels across labeled cell clusters.

size. In the strict final comparison, the robust panel had nominal macro MAE 0.0606 and ROC AUC 0.9860, compared with MAE 0.0613 and ROC AUC 0.9865 for nominal-only selection. Across the seven tested holdout regimes, macro worst-case MAE was 0.1103 for robust selection and 0.1196 for nominal-only selection. In the combined regime, robust selection had MAE 0.1103 and ROC AUC 0.9321, compared with MAE 0.1196 and ROC AUC 0.9044 for nominal-only selection. Thus, nuisance-aware selection yielded lower quantitative error in this simulation-based comparison, but not uniformly better detection metrics.

The findings are limited to pseudobulks generated from the supplied single-cell dataset, the specified nuisance ranges, and the fixed discovery-holdout protocol. The experiments provide no uncertainty estimates or statistical tests, use no independent biological samples, and evaluate an expression-based normalized score rather than directly measured qPCR cycle-threshold values. The results therefore do not establish transfer to laboratory assays or performance under untested nuisance distributions.

Future work should evaluate the selected panels on independent biological samples and experimentally measured qPCR data with technical replicates and assay-specific efficiency estimates. Further studies should test empirically calibrated nuisance distributions, alternative normalization strategies, more

Held-out cell-type detection performance — run_5

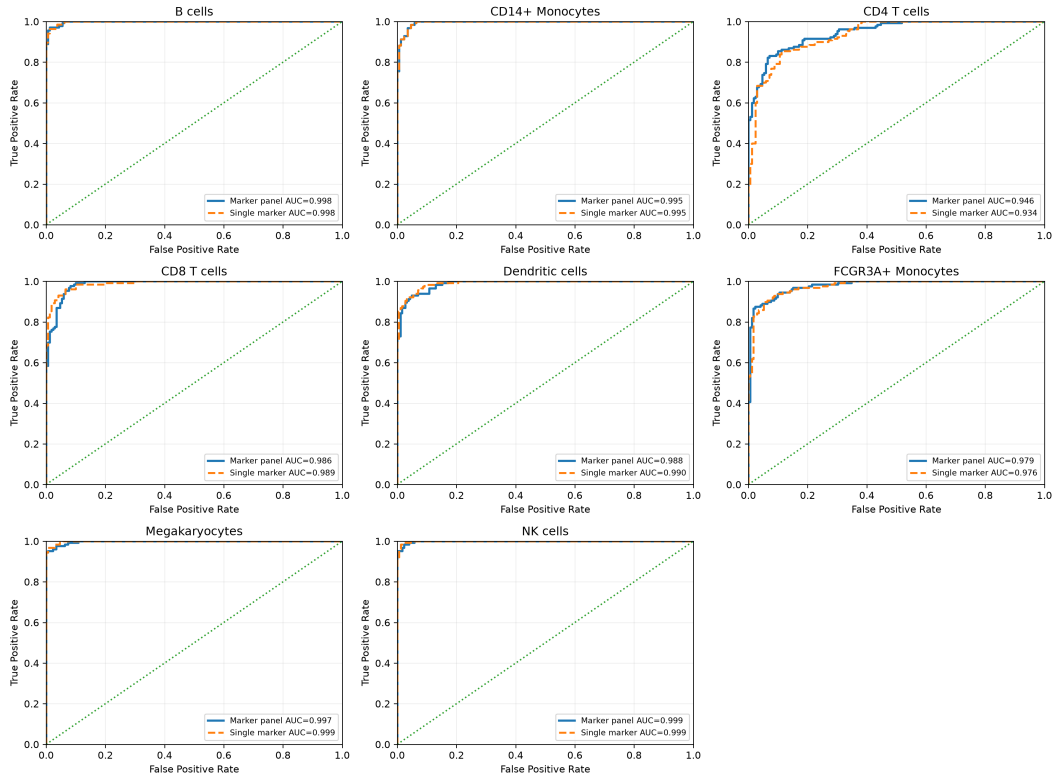


Figure 4: Cluster-wise nominal holdout ROC curves for the selected multi-gene panels and their best single-marker comparators.

diverse cell populations, and external datasets collected under different protocols. These experiments can determine whether the observed reduction in simulated quantitative error transfers to laboratory measurements and can clarify how panel size should be balanced against accuracy, calibration, and measured assay variation.

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Quantitative cell-fraction prediction — run_5

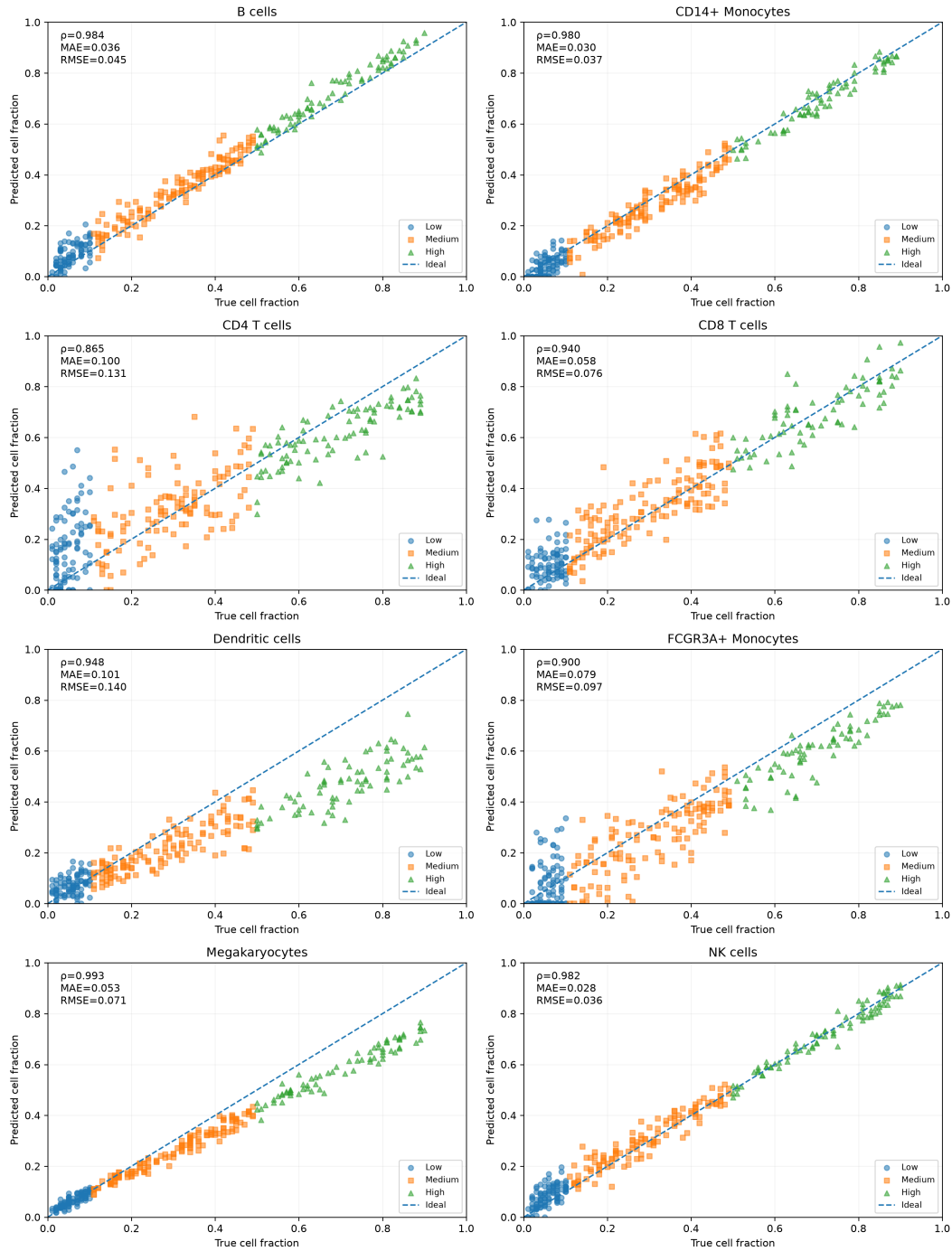


Figure 5: Nominal holdout predicted-versus-true target fractions for the corrected selection comparison. Points use distinct markers for low, medium, and high true-fraction bins; the diagonal denotes ideal prediction.

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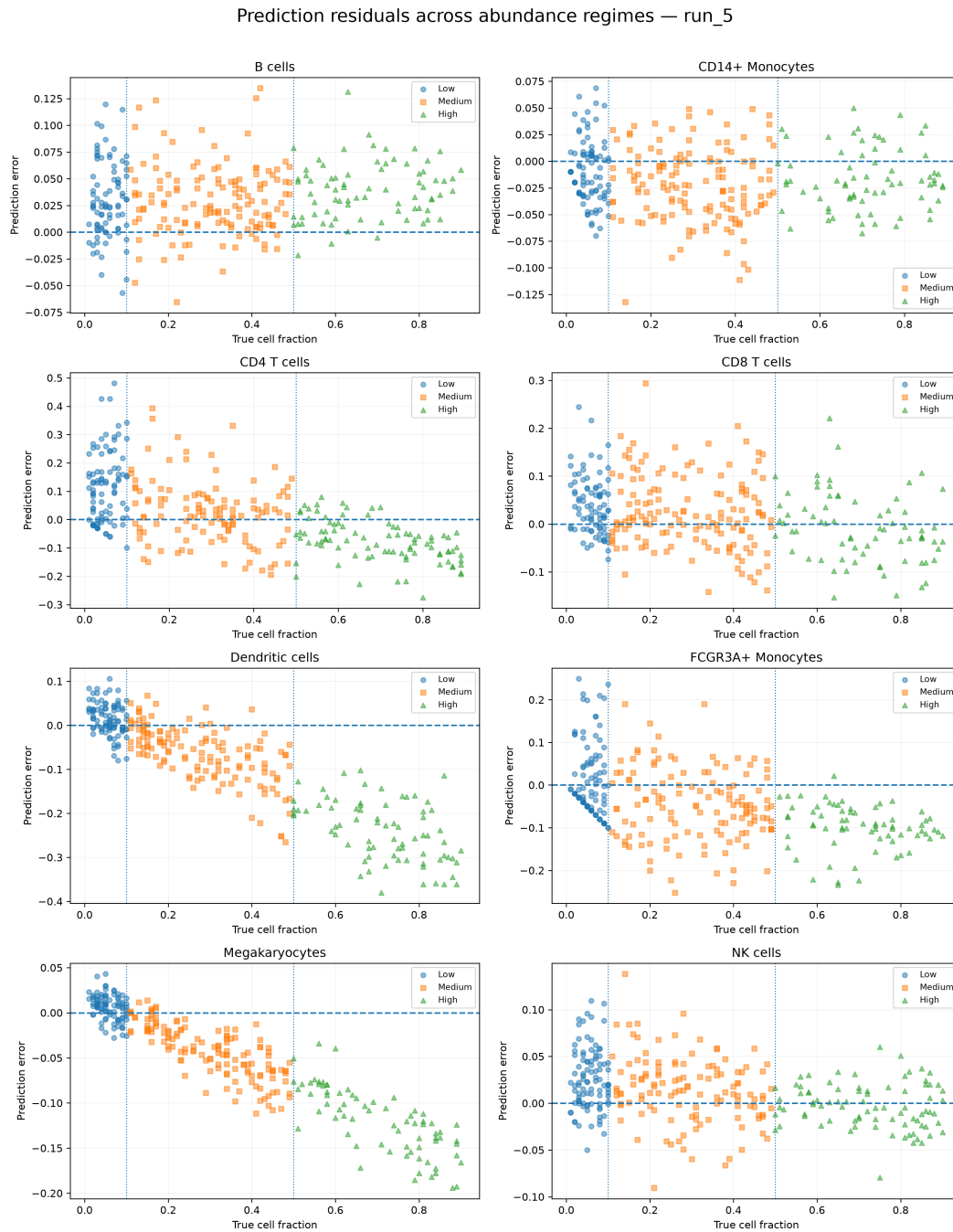


Figure 6: Nominal holdout prediction residuals, defined as predicted minus true fraction, across target clusters and abundance bins. Vertical guides mark the low- and high-fraction thresholds.

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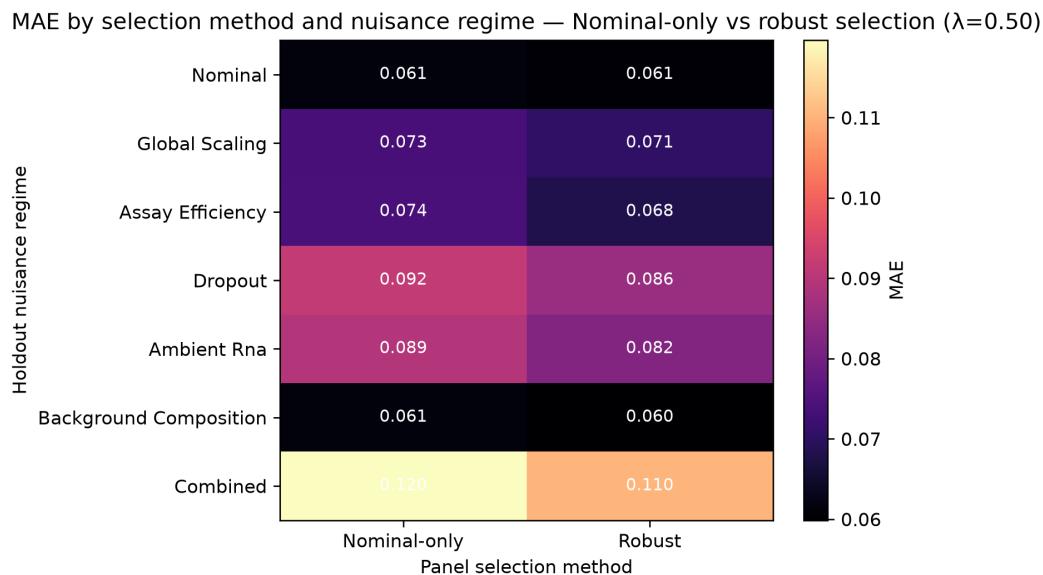


Figure 7: Macro MAE for nominal-only and nuisance-robust selection across the nominal and simulated nuisance holdout regimes. Cell annotations report regime-specific macro MAE.

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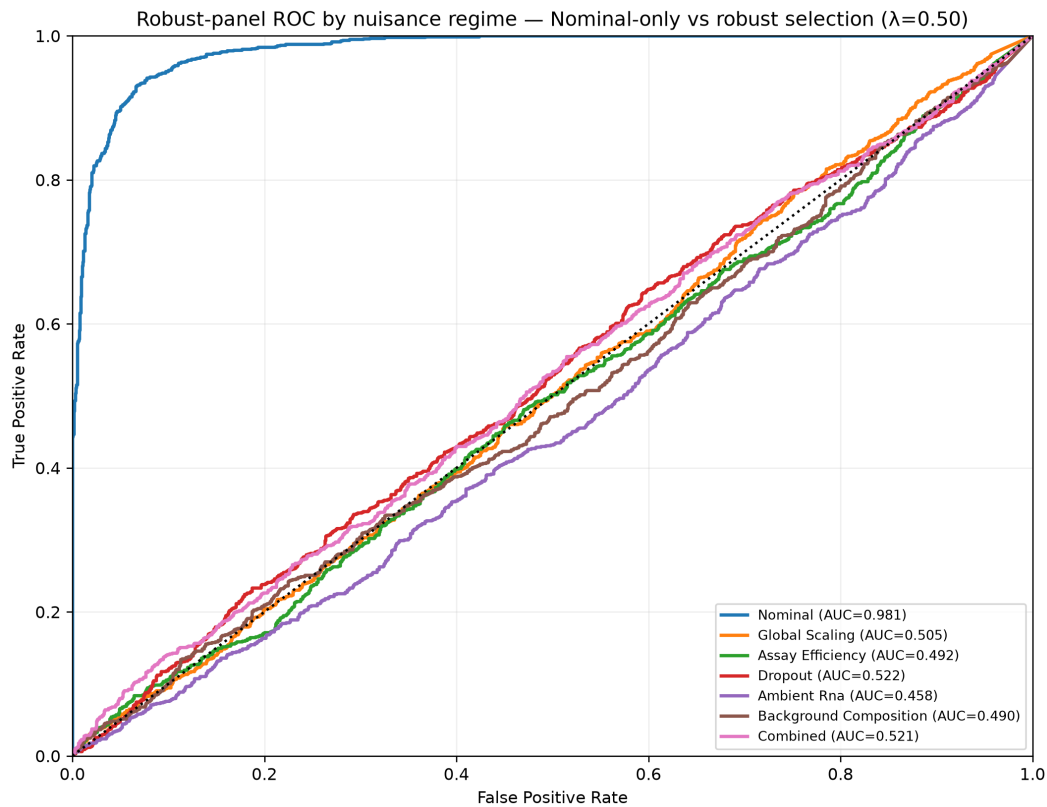


Figure 8: Aggregate ROC curves for the robust panel across nominal and simulated nuisance holdout regimes in the final protocol. Each curve corresponds to one regime and is labeled with its ROC AUC.

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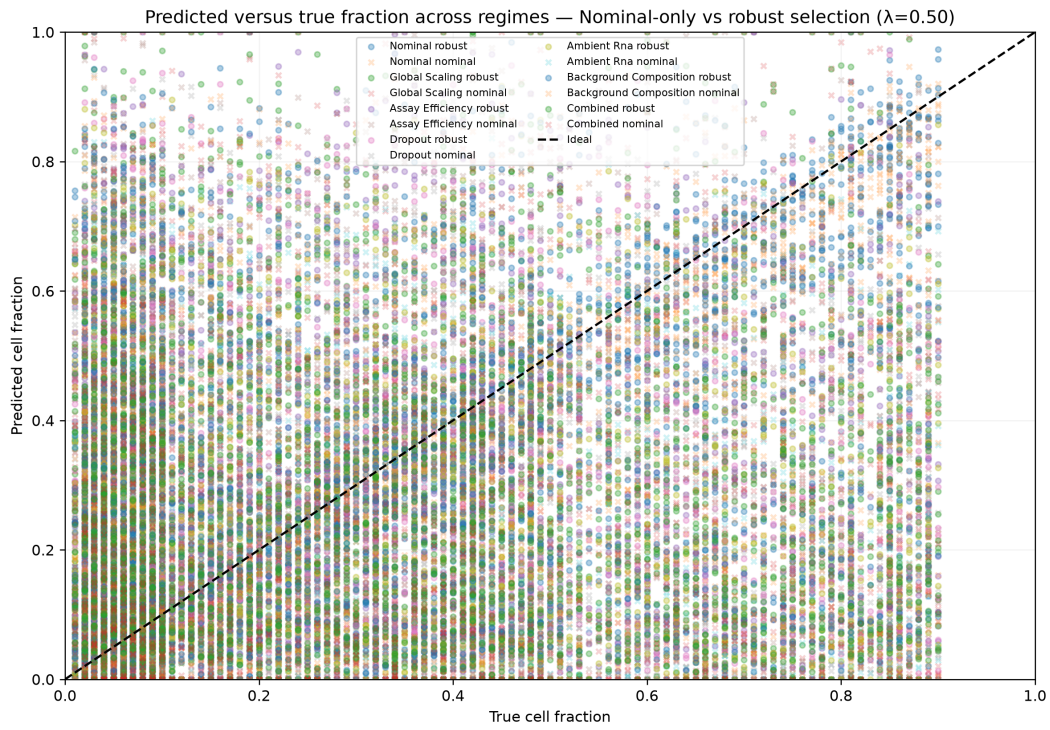


Figure 9: Predicted-versus-true target fractions across holdout nuisance regimes for nominal-only and robust selection in the final protocol. The diagonal denotes ideal prediction.